

Phase 1

1. Vendor - Project Initiation & Document Submission Mark work as completed Upload all required documents (COC, Photos, Permits, Receipts, Safety Forms, Drawings) View SLA countdown (7-day deadline) Receive confirmation numbers Color-coded alerts for overdue projects
2. Supervisor - Project Creation & Assignment Create new projects Assign to vendors Set SLA deadlines Approve submitted documents View project statistics dashboard Priority-based project management
3. Clerk - Document Validation Validate document submissions Check file formats and quality Flag quality issues (file size, type validation) Generate confirmation numbers Send notifications to vendors and supervisors Approve/reject individual documents
4. Admin - User Access Management Create/manage user accounts Assign roles and permissions Activate/deactivate users View user directory Monitor system security Track user sessions

Phase 2

Vendor - Penalty Notifications & Disputes View penalty details with amount breakdowns Submit disputes with detailed explanations Track penalty status (Draft, Issued, Paid, Waived, Disputed) See summary cards and penalty history

Clerk - Penalty Documentation Support View delayed projects requiring documentation Auto-calculate penalties (delay days $\times$ 0.1% per day) Prepare penalty memos with checklist Forward documentation to supervisor

Supervisor - Penalty Review & Approval Color-coded dashboard (■ GREEN / ■ YELLOW / ■ RED status) Review pending penalties Approve/reject with notes Monitor real-time project status

Team Leader - Secondary Approval & Consistency Review high-value penalties ($\geq$ ■50,000) Secondary approval workflow Department-wide consistency monitoring Supervisor performance comparison Analytics dashboard

Sector Manager - Sector-Wide Compliance Sector-wide compliance dashboard Monitor 150+ projects and ■25M pipeline Vendor compliance performance tracking Strategic reports and trends Multi-department oversight

Phase 3

1. Supervisor QI Assignment Dashboard AI-powered QI recommendations (workload + GPS + specialization) QI team capacity overview with real-time percentages Pending projects awaiting assignment Accept or override AI suggestions Track recent assignments
2. Team Leader Workload Balancing Multi-supervisor aggregate view Department-wide QI capacity monitoring Auto-balance algorithm ($\pm 8\%$ variance target) Workload distribution by supervisor Variance tracking and optimization
3. QI Mobile Dashboard Inspection queue sorted by urgency (URGENT/DUE/SCHEDULED) Accept or request reassignment View assignment details Capacity tracker display Push notifications integration
4. Clerk QI Coordination Process reassignment requests QI calendar management (7-day view) Auto-calendar integration Email + push notifications Schedule conflict resolution

Phase 4

1. QI Digital Inspection Interface GPS auto-timestamp on arrival Project-specific digital checklists (Electrical, Civil, Mechanical) PASS/FAIL marking with photo capture Voice-to-text notes Offline mode with sync capability Digital signature Auto-generates defect reports when items fail
2. Vendor Defect Corrections Interface View all defect reports with failed items Upload corrective action photos Submit correction notes Track deadline countdown Request deadline extensions Handle multiple failure attempts Escalation warnings after 3 failures
3. Supervisor Inspection Monitoring Real-time inspection statistics dashboard View all inspection records with filters Approve/reject vendor corrections Escalate cases after 3 failures to Team Leader Monitor re-inspection schedule View detailed inspection reports with photos
4. Clerk Re-Inspection Scheduling Schedule re-inspections with same QI Coordinate QI calendar Auto-notify QI and vendor Reschedule inspections Archive inspection documents (10-year retention) Track focus items (previously failed items)
5. Engineering Aide Coordination Dashboard Coordinate re-inspection logistics Technical review of complex failures Escalate to higher authority Monitor active escalations Track re-inspection schedule Technical evaluation and recommendations

Phase 5

1. Clerk - Pre-Billing Validation 6-point validation checklist (inspection approved, documents complete, no pending corrections, penalties calculated, account good standing, no legal holds) Document verification (COC, QI Report, receipts, penalty memos) Submit for billing workflow Real-time validation status
2. Supervisor - Billing Calculation & Approval Billing calculation breakdown (contract amount - penalties - deductions = net amount) Penalty details display Invoice number generation Approve &

generate invoice workflow Auto-updates project status to "Awaiting Payment"

3. Team Leader - Financial Oversight High-value invoice review ($\geq$ \$100,000 threshold) Financial metrics dashboard (pending review, total value, high-value count) Value categorization (Very High $\geq$ \$1M, High $\geq$ \$500K, Medium $\geq$ \$100K) Route to Finance with approval

4. Vendor - Invoice & Payment Tracking Personal invoice dashboard Payment status tracking (Unpaid, Partially Paid, Paid, Overdue) Invoice details view PDF download capability Performance stats (total invoices, pending payment, YTD collected)

5. Sector Manager - Billing Metrics Sector-wide billing analytics Key metrics (total billed, collected, pending, avg payment cycle) On-time billing rate tracking (target: 95%) Collection efficiency metrics Monthly billing trends visualization Report export functionality

Phase 6 1. Clerk - Pre-Billing Validation Validates 6 critical checks before billing Document completeness verification Inspection approval confirmation Penalty calculation validation Account status checks Submits validated projects for billing approval

2. Supervisor - Billing Approval Reviews billing calculations Displays contract amount minus penalties Shows detailed penalty breakdown Approves and generates invoices Routes to Finance with full documentation

3. Vendor - Payment & Invoice Tracking Monitors all invoices and payment status Downloads invoices and payment receipts Views performance scorecard (on-time rate, quality, SLA compliance) Tracks payment cycles (Net 30 days) Color-coded status indicators

4. Team Leader - Billing Oversight Department-wide billing dashboard Reviews high-value invoices ($>$ \$100,000) Monitors total pending billing value Tracks average payment cycle times Routes approved invoices to Finance

5. Sector Manager - Billing Metrics & Portfolio Sector-wide \$25M+ pipeline overview Vendor performance rankings with renewal recommendations 6-month billing trends visualization Strategic partnership approvals Budget vs. actuals tracking

6. Admin - Document Archiving & Audit 10-year retention compliance management Cloud + local backup tracking Complete audit trail maintenance Read-only permissions enforcement Storage capacity monitoring